Chapter 8: WKB Approximation

Quantum Mechanics

1 Tunneling: Intuition and Terminology

Remark 1.1 (Physical Picture of Tunneling). [rem:tunneling-intuition] **TODO**

Convention 1.1 (Classically Allowed Region). [conv:allowed-region] For fixed E and V, the positions x satisfying $E > V(x)$ form the *classically allowed region*

Convention 1.2 (Classically Forbidden Region). [conv:forbidden-region] For fixed E and V, the positions x satisfying $E < V(x)$ form the *classically forbidden region*

Definition 1.1 (Transmission Coefficient). [def:transmission-coefficient] **TODO**

2 WKB Wavefunctions in the Allowed Region

Definition 2.1 (Classical Momentum Function). [def:classical-momentum] **TODO**

Hypothesis 2.1 (Slowly Varying Amplitude). [hypo:slowly-varying-amplitude] ***TODO***

Theorem 2.1 (Allowed-Region WKB Form). [thm:wkb-allowed-region] ***TODO***

Proof. **TODO** □

Remark 2.1 (Validity of the WKB Approximation). [rem:wkb-validity] **TODO**

Theorem 2.2 (Quantization Between Two Infinite Walls). [thm:hard-wall-quantization] ***TODO***

Proof. **TODO** □

3 WKB Wavefunctions in the Forbidden Region

Definition 3.1 (Forbidden-Region Momentum Magnitude). [def:forbidden-momentum] **TODO**

Theorem 3.1 (Forbidden-Region WKB Form). [thm:wkb-forbidden-region] ***TODO***

Proof. **TODO** □

Definition 3.2 (Barrier Action). [def:barrier-action] **TODO**

Theorem 3.2 (Leading WKB Tunneling Probability). [thm:wkb-transmission] ***TODO***

Proof. **TODO** □

Remark 3.1 (Abrupt Boundaries and Transmission Prefactors). [rem:transmission-prefactors] **TODO**

4 Gamow's Theory of Alpha Decay

Remark 4.1 (Alpha-Decay Barrier Model). [rem:alpha-decay-barrier] **TODO**

Theorem 4.1 (Barrier Penetration in Alpha Decay). [thm:alpha-decay-penetration] ***TODO***

Proof. **TODO** □

Remark 4.2 (Escape Attempts and Decay Rate). [rem:alpha-decay-attempts] **TODO**

Theorem 4.2 (Gamow's Estimate of the Decay Lifetime). [thm:gamow-lifetime] ***TODO***

Proof. **TODO** □

5 Turning Points and Connection Formulas

Definition 5.1 (Simple Turning Point)**.** [def:simple-turning-point] **TODO**

Remark 5.1 (Airy Functions and Overlap Matching). [rem:airy-matching] **TODO**

Theorem 5.1 (Upward-Sloping Connection Formula)**.** [thm:upward-connection-formula] ***TODO***

Proof. **TODO** □

Theorem 5.2 (Downward-Sloping Connection Formula)**.** [thm:downward-connection-formula] ***TODO***

Proof. **TODO** □